AI JOURNEY • MODUL 01 YAKUNI • PORTFOLIO LOYIHASI 01

CLI Student Management System (Talabalar Boshqaruv Tizimi)

Python asoslari, OOP, JSON Persistence va CLI orqali to'liq ishlaydigan portfolioga loyiq konsol tizimi

Status: Yakunlangan (Production Ready) **Arxitektura:** 4 Qatlamli Modulli Tizim **Dasturlash tili:** Pure Python 3.10+

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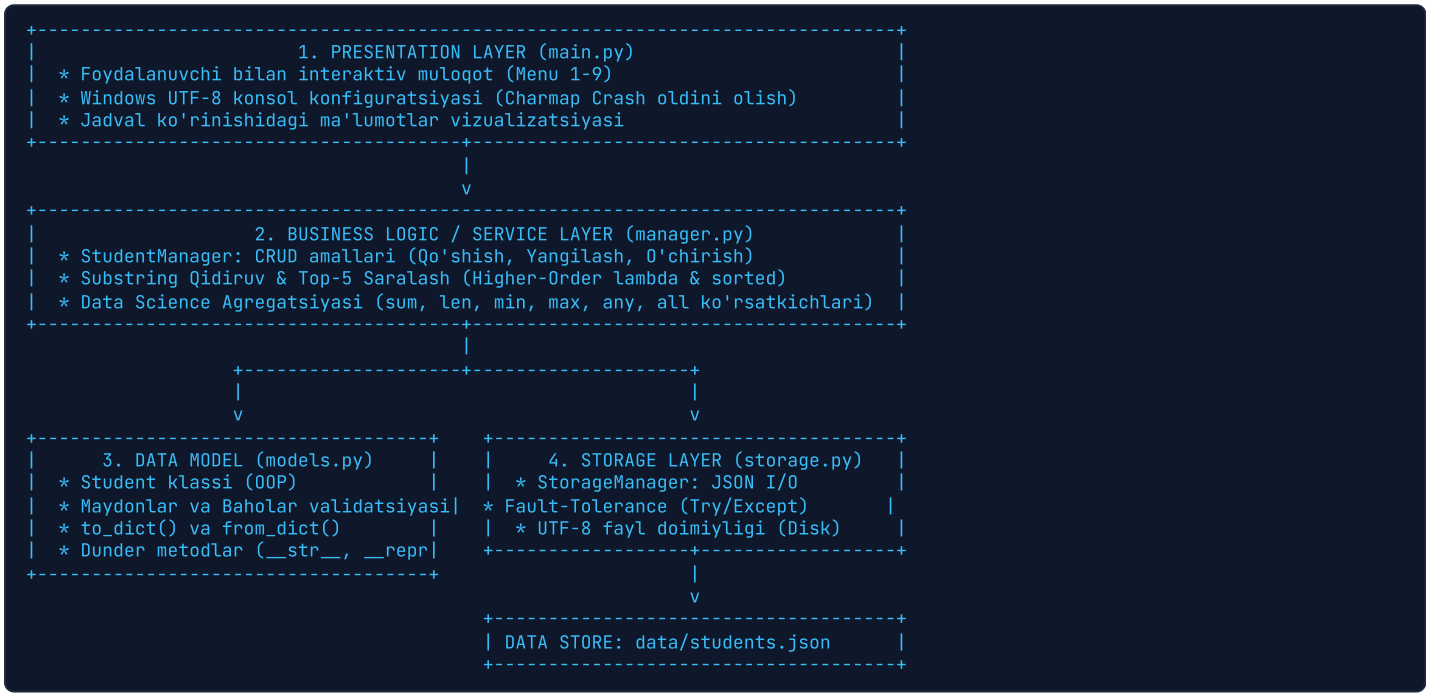
1. Loyihaning Maqsadi va Umumiy Konsepsiyasi

Dasturlash va Sun'iy intellekt (AI) muhandisligida muvaffaqiyatga erishish uchun faqat darslardagi alohida sintaksis mashqlarini bajarish yetarli emas. Haqiqiy dasturchi o'rganilgan barcha mavzularni — *o'zgaruvchilar, shartlar, sikllar, funksiyalar, ma'lumot tuzilmalari, fayllar, istisnolar (Exceptions) va OOP tamoyillarini* — bitta mustahkam, professional arxitekturada birlashtira olishi kerak.

CLI Student Management System — bu o'quv markazlari yoki universitetlar uchun talabalarni ro'yxatga olish, ularning imtihon ballarini hisoblash, guruh o'rtacha GPA ko'rsatkichini aniqlash, Top-5 a'lochilarni saralash va barcha ma'lumotlarni qattiq diskdagi JSON bazada xavfsiz saqlashni ta'minlovchi to'liq mustaqil konsol platformasidir.

2. Tizim Arxitekturasi (4 Qatlamli Tizim)

Loyiha professional dasturiy ta'minot standartlariga binoan **Separation of Concerns** (Vazifalarni qat'iy taqsimlash) qoidasiga asoslangan 4 qatlamdan iborat:



3. Loyiha Papkalar Strukturasi va Rol Taqsimoti

Separation of Concerns (Mas'uliyatlar taqsimoti): Loyiha kodlari bitta katta faylga yozilmasdan, 4 ta mustaqil modulga ajratilgan. Bu kodni o'qish, testlash va kelajakda kengaytirishni osonlashtiradi:

- `src/models.py` — Talaba sub'ektining atributlari, baholash logikasi va ma'lumotlar validatsiyasi.
- `src/storage.py` — Xotiradagi ob'ektlarni diskdagi JSON fayliga yozish (Serialization) va o'qish (Deserialization).
- `src/manager.py` — Barcha talabalar to'plamini boshqarish, saralash, qidirish va guruh statistikasini tahlil qilish.
- `src/main.py` — Foydalanuvchi uchun qulay menyu, xatolardan himoyalangan konsol kiritish/chiqarish (I/O).
- `src/data/students.json` — Dastur o'chirilganda ham ma'lumotlar saqlanib qoladigan doimiy ma'lumotlar bazasi.

4. Ma'lumotlar Modeli: models.py (OOP va Validatsiya)

Tizim markazida `Student` sinfi turadi. U talabaning shaxsiy ma'lumotlarini o'zida birlashtiradi va ma'lumotlar yaxlitligini (Data Integrity) qat'iy tekshiradi.

Talaba Atributlari va Tiplari:

- `student_id: int` — Unikal ketma-ket raqam
- `full_name: str` — To'liq ism (Titlecase)
- `email: str` — Unikal elektron pochta
- `grades: list[float]` — Imtihon ballari (0-100)
- `enrolled_date: str` — Sana (YYYY-MM-DD)

Validatsiya Qoidalari (_validate):

- Ism bo'sh bo'lmasligi va kamida 3 belgi
- Email ichida `@` va `.` mavjudligi
- Har bir baho `0 ⩽ baho ⩽ 100`
- Buzilsa `ValueError` chiqariladi

```
class Student:
    def __init__(self, student_id: int, full_name: str, email: str, grades: list = None):
        self.student_id = student_id
        self.full_name = full_name.strip().title()
        self.email = email.strip().lower()
        self.grades = grades if grades is not None else []
        self._validate()

    def average_grade(self) -> float:
        if not self.grades: return 0.0
        return round(sum(self.grades) / len(self.grades), 2)

    def academic_status(self) -> str:
        avg = self.average_grade()
        if not self.grades: return "Baholanmagan"
        elif avg ⩾ 86.0: return "A'lochi (Honors) 🌟"
        elif avg ⩾ 71.0: return "Yaxshi (Good) 👍"
        elif avg ⩾ 55.0: return "Qoniqarli (Passing) ⚠️"
        else: return "Qarzdor (Failing) ❌"
```

5. Fayl Tizimi va Doimiylik: storage.py (JSON Persistence)

Operativ xotiradagi (RAM) Python ob'ektlari dastur yopilganda o'chib ketadi. `StorageManager` ma'lumotlarni qattiq diskdagi JSON formatiga o'tkazish (Serialization) va qayta tiklash (Deserialization) xizmatini beradi.

Fayl Yo'lini Xavfsiz Aniqlash (Absolute Path Resolution):

Dastur terminalda qayerdan ishga tushirilishidan qat'i nazar xato qilmasligi uchun: `base_dir = os.path.dirname(os.path.abspath(__file__))` va `os.path.join(base_dir, "data", "students.json")` qo'llaniladi.

```
class StorageManager:
    def load_students(self) -> list[Student]:
        if not os.path.exists(self.file_path):
            return []  # Fayl hali yo'q bo'lsa bo'sh ro'yxat qaytaradi
        try:
            with open(self.file_path, mode="r", encoding="utf-8") as f:
                data = json.load(f)
                return [Student.from_dict(item) for item in data]
        except (json.JSONDecodeError, Exception) as e:
            print(f"⚠️ Ogohlantirish: JSON o'qishda xatolik: {e}")
            return []

    def save_students(self, students: list[Student]) -> bool:
        try:
            data = [s.to_dict() for s in students]
            with open(self.file_path, mode="w", encoding="utf-8") as f:
                json.dump(data, f, indent=2, ensure_ascii=False)
            return True
        except Exception as e:
            return False
```

6. Biznes Mantiq va Tahlil Dvigateli: manager.py

`StudentManager` sinfi servis qatlami (Service Layer) hisoblanadi. U talabalar ro'yxatini xotirada keshlaydi, CRUD amallarini bajaradi va ma'lumotlar o'zgarganda ularni avtomatik tarzda `StorageManager` orqali diskka saqlaydi.

CRUD & Boshqaruv:

- `add_student()` — Dublikat emailni tekshiradi, keyingi bo'sh ID ni hisoblaydi (`max(id) + 1`).
- `search_students()` — Katta-kichik harfni ajratmasdan ism yoki emaildan qidiradi (Substring).
- `update_student()` — Ism yoki emailni qayta tekshirgan holda tahrirlaydi.
- `delete_student()` — Talabani ro'yxatdan o'chirib diskni yangilaydi.

Tahliliy Funksiyalar (Analytics):

- `get_top_students(top_n=5)` — Lambda ifodasi yordamida eng a'lochi talabalar reytingi.
- `get_statistics()` — Guruhning umumiy GPA ko'rsatkichi, eng yuqori va past baholar.
- `all() va any()` — Guruhda qarzdorlar borligini yoki hamma talabalar imtihondan o'tganligini aniqlash.

```python
def get_top_students(self, top_n: int = 5) -> list[Student]:
    graded = [s for s in self.students if s.grades]
    return sorted(graded, key=lambda s: s.average_grade(), reverse=True)[:top_n]

def get_statistics(self) -> dict:
    graded = [s for s in self.students if s.grades]
    averages = [s.average_grade() for s in graded]
    return {
        "total_students": len(self.students),
        "class_average": round(sum(averages) / len(averages), 2) if averages else 0.0,
        "highest_avg": max(averages) if averages else 0.0,
        "lowest_avg": min(averages) if averages else 0.0,
        "honors_count": len([s for s in graded if s.average_grade() ⩾ 86.0]),
        "has_failures": any(s.average_grade() < 55.0 for s in graded),
        "all_passing": all(s.average_grade() ⩾ 55.0 for s in graded) if graded else False
    }
```

7. Konsol Foydalanuvchi Interfeysi: main.py

`main.py` foydalanuvchi bilan muloqot qiluvchi konsol boshqaruv markazidir. U qulay menyu, chiroyli jadvallar va kiritilgan ma'lumotlar xavfsizligini ta'minlaydi.

Windows UTF-8 Muammosini Hal Qilish (Crucial Fix):

Windows PowerShell yoki CMD standart holatda `cp1252` da ishlaydi. Konsolda emoji yoki maxsus belgilar (☀️ , ❌ , 👍) chiqqanda dastur `UnicodeEncodeError` bilan qulamasligi uchun skript boshida: `if hasattr(sys.stdout, 'reconfigure'): sys.stdout.reconfigure(encoding='utf-8')` buyrug'i qo'shildi.

```python
def print_student_table(students: list):
    if not students:
        print("\nℹ️ Hozircha tizimda talabalar mavjud emas.")
        return
    print("\n" + "-" * 75)
    print(f"{'ID':<5} | {'F.I.SH.':<22} | {'EMAIL':<22} | {'O\'RT.':<6} | {'STATUS':<15}")
    print("-" * 75)
    for s in students:
        print(f"{s.student_id:<5} | {s.full_name:<22} | {s.email:<22} | {s.average_grade():<6.1f} | {s.academic_status():<15}
    print("-" * 75)
```

8. Modul 01 Mavzularining Loyihadagi Xaritasi (Sintez)

Ushbu jadval 1-moduldagi barcha 11 ta nazariy va amaliy dars qanday qilib ushbu loyihaga organik ravishda tatbiq etilganini ko'rsatadi:

Dars Kodi	Dars Mavzusi	Loyihadagi Amaliy Qo'llanilishi
M01-L02	Variables & Data Types	`student_id (int)`, `full_name (str)`, `grades (list[float])`, `enrolled_date (str)` tiplari.
M01-L03	Operators & I/O	Matnlarni tekislab formatlash (`f-strings` kengliklari), `input()` orqali ma'lumot olish.
M01-L04	Conditionals (if/elif/else)	Talabaning akademik holatini aniqlash (Honors, Passing, Failing), tasdiqlash so'rovlari.
M01-L05	Loops (for, while)	Boshqaruv menyusi uchun cheksiz `while True` sikli, talabalar va baholarni `for` bilan aylanish.
M01-L06	Functions & Scope	Modulli UI funksiyalari, parametrlarni uzatish, lambda orqali saralash kaliti.
M01-L07	List, Tuple, Set, Dict	Talabalar ro'yxati (`list`), JSON yozuvlari (`dict`), tahliliy to'plamlar.
M01-L08	Strings & Built-ins	`.strip()`, `.title()`, `.lower()`, `sum()`, `len()`, `max()`, `min()`, `all()`, `any()`.
M01-L09	File Handling (JSON)	`json.dump()` va `json.load()` yordamida ma'lumotlarni qattiq diskka doimiy saqlash.
M01-L10	Exception Handling	`try/except ValueError`, `JSONDecodeError` va nolga bo'linish xatolarini ushlash.
M01-L11	OOP Asoslari	`Student` va `StorageManager` sinflari, inkapsulyatsiya, `self`, `@classmethod`.

9. Konsol Ishchi Natijalari (Terminal Demonstration)

```
================================================================
   🎓  TALABALARNI BOSHQARISH TIZIMI (CLI STUDENT SYSTEM)
================================================================
  1. 📋 Barcha talabalarni ko'rish     5. 🖍️  Talaba ma'lumotlarini yangilash
  2. ➕ Yangi talaba qo'shish          6. 🗑️  Talabani o'chirish
  3. 📝 Talabaga baho qo'shish         7. 🏆 Top talabalar reytingi (Top 5)
  4. 🔍 Talabani qidirish              8. 📊 Umumiy statistika va hisobot
                                       9. 🚪 Dasturdan chiqish
👉 Tanlovingizni kiriting (1-9): 8

================================================================
   📌  TIZIM STATISTIKASI & HISOBOT
================================================================
📊 Jami talabalar soni:         3     | 🌟 A'lochilar soni (≥86):    1
📝 Baholangan talabalar:        3     | 👍 O'tish ballini olganlar:   1
📈 Guruh o'rtacha balli:        73.56 | ❌ Qarzdorlar soni (<55):      1
🥇 Eng yuqori o'rtacha ball:    91.67 | ⚠️ Qarzdorlar mavjudmi:       Ha
📉 Eng past o'rtacha ball:      49.0  | 🎯 Hamma imtihondan o'tdimi:  Yo'q
```

10. Modul Xulosasi va Keyingi Bosqich

🎉 **Modul 01: Python Fundamentals Muvaffaqiyatli Yakunlandi!**

Siz Python dasturlash tilining barcha bazaviy poydevorini o'rganib, ilk mustaqil portfolio loyihangizni to'liq ishlab chiqdingiz. Endi kompyuteringizda real ishlaydigan, xatolardan himoyalangan va toza arxitekturaga ega tizim mavjud.

🚀 **Keyingi bekat — Modul 02: Mathematics & Statistics for AI!**

Kelgusi modulda biz Sun'iy intellekt va Machine Learning algoritmlarining yuragi bo'lgan Ehtimollar nazariyasi, Matritsalar, Vektorlar, Hosilalar va Gradient Descent matematikasini intuitiv va vizual tarzda o'rganishni boshlaymiz.